MILITARY SPECIFICATION

BAR, ORTHOPEDIC FRAME, TURNTABLE

1. SCOPE

1.1 This specification covers one type of telescopic, turntable suitable for use with an overhead frame on hospital beds.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

Fed. Test Method Std. No. 151 Metals, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6530

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DRAWING

Defense Medical Supply Center

20137 BAR, ORTHOPEDIC FRAME, TURNTABLE

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The components of the bar shall be fabricated from material as specified:

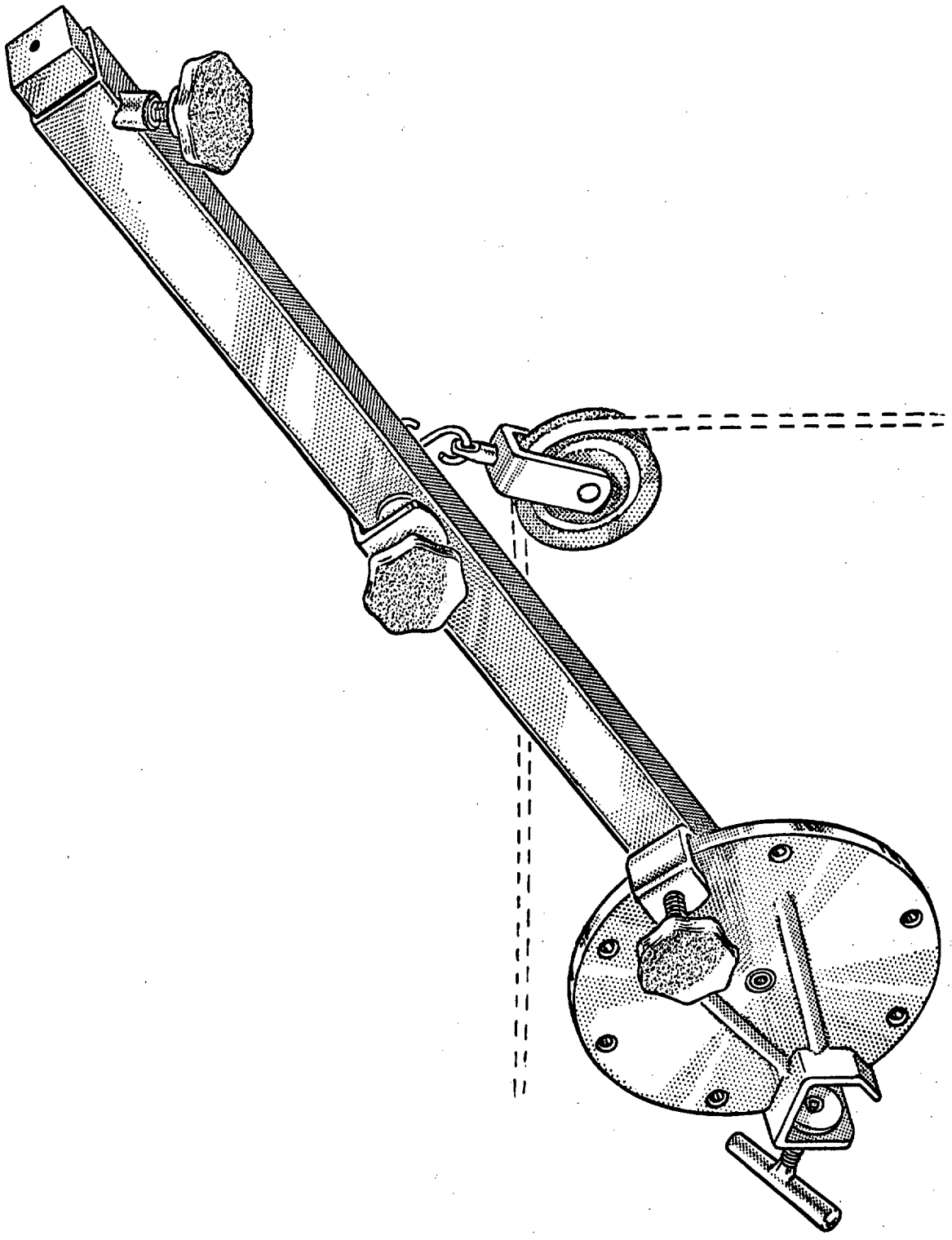
Component	Material
Tubing	Cold-rolled SAE 1010 steel
Turntable	Hot-rolled steel plate
Clamps	Rail carbon steel or SAE 1035 steel
Handles and screws	Round, cold-rolled, mild steel
Hand wheel	Cast iron
Swivel pulley and clamp	
Clamp wheel	Cast iron
Clamp stud	Cold-rolled steel
"C" clamp	Rolling mill flat or rerolled rail stock
Clamp plate	Hot-rolled steel
Swivel pin	Steel wire
Pulley yoke, swivel pin	Steel wire
Pulley yoke	Rolling mill stock
Pulley	0.042 inch (No. 19, United States Standard Revised) sheet

3.2 Construction. The bar shall be in strict accordance with Drawing 20137 and Figure 1. The bar shall have a pivot arc of 270° and shall be capable of being locked at seven points by means of the hand screw clamp. A stop shall be provided to prevent the inner square telescoping tube from slipping out of the outer tube when in the extended position.

3.2.1 Clamp and pulley. One loose swivel pulley and clamp shall be provided with each bar. Clamp and pulley shall be in accordance with Figures 2, 3, and 4. Dimensional tolerances shall not be cumulative.

3.3 Load requirements. The assembled components shall be capable of safely supporting a minimum weight load of 75 pounds when tested as specified in 4.4.2.

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BAR, ORTHOPEDIC FRAME

Figure 1

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3.4 Finish. The entire bar assembly shall receive a cadmium-or bright zinc-plated finish, followed by a baked-on coat of clear lacquer or a coat of baked-on clear synthetic-type varnish. The clamp and pulley shall be zinc plated. Handwheels shall be plastic coated.

3.5 Identification marking. Each bar shall be legibly and permanently marked in an inconspicuous location with the following information:

Federal stock number.
Manufacturer's name or registered trademark.
Contract or purchase order number.
U.S.

Marking secured by adhesives shall not be used.

3.6 Workmanship. Workmanship shall be first class throughout. Bars shall be free from defects which detract from their appearance or may impair their serviceability.

3.6.1 Welding. The welds shall be thorough, nonporous, and free of pits, cracks, and other imperfections. Welding shall be continuous, as shown on the drawing, and shall be thoroughly cleaned to remove flux, spatter, scale and discoloration before plating.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total (major and minor combined)defects	II	4.0
For dimensional and weight examination	S-3	2.5

4.2.2 For test. Sampling for load test shall be conducted in accordance with MIL-STD-105. Inspection level shall be S-3, and acceptance number shall be zero regardless of lot size.

4.2.3 Raw materials. Four ounces of each lot of metal used in the construction of the bar shall be selected for chemical analysis.

4.3 Examination. The bars shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Bar not complete with all specified component parts.
102 Inner tube does not move smoothly within outer tube.
103 Stop does not prevent inadvertent removal of inner tube.
104 Threaded parts do not operate smoothly.
105 Component parts and welds in assembled bar not free of cracks, tears, and ruptures.
106 Plated and painted surfaces not free of peeled, chipped, and corroded areas.
107 Exposed edges not free of burs and sharpness.

Minor

201 Identification marking not complete, permanent, and legible.
202 Finished surfaces not free of scratches, pits and grind marks.
203 Welds not free from flux, discoloration, and splatter before plating.

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensional examination. The bars shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Fed. Test Method Std. No. 151.

4.4.2 Load test. A rigid fixture of unspecified design having a 1-1/2 inch square bar shall be supplied to simulate an overhead frame on a hospital bed. The orthopedic bar shall be clamped in place by tightening the hand wheel. The outer tube of the orthopedic bar shall be pivoted and shall be locked at the middle hole of the disc by means of the hand screw clamp. The inner telescoping tube of the orthopedic bar shall be extended to its maximum length and locked in place with the hand wheel. The swivel pulley shall be clamped to the middle of the inner telescoping tube and tightened by means of the hand screw clamp. A rope shall be supplied which shall have a diameter of 1/8-inch and shall be of suitable length for the test. The rope shall be properly threaded through the pulley. One end of the rope shall be fixed. The other end of the rope shall freely support a 75 pound weight. The load shall be freely suspended below the level of the turntable disc. The bar shall be observed for evidence of failure.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One(1) bar, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I - Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent spillage of contents under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall be in accordance with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with applicable paragraphs of appendix thereto.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; or PPP-B-621, table I; or PPP-B-636, type CF, class domestic (Method I closure shall be utilized)

5.3 Marking.

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5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing. (see 5.1 and 5.2)

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

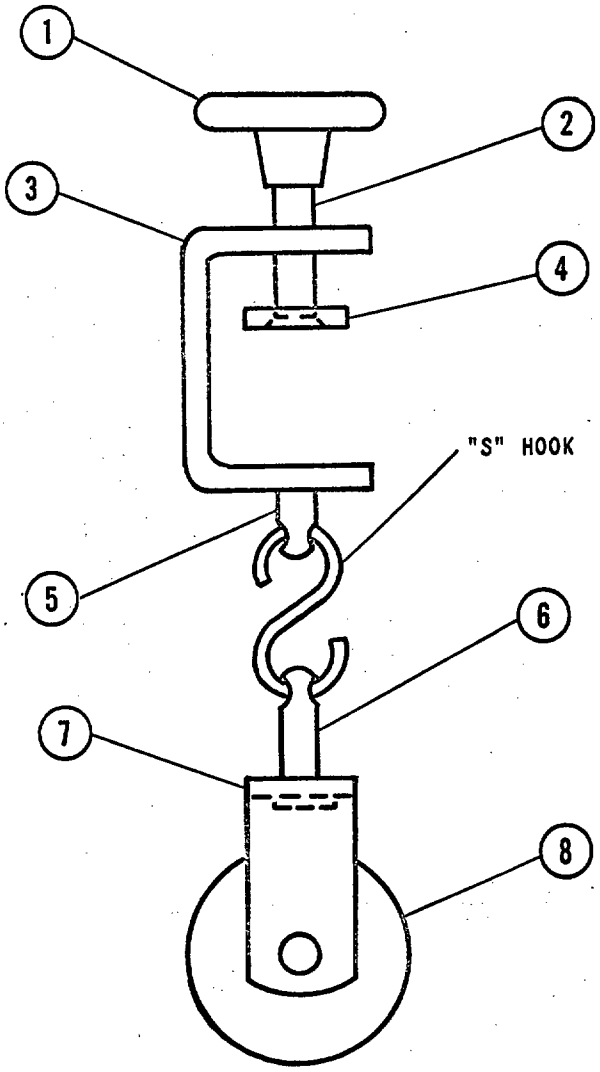
Custodians:
Army- MD
Navy- MS
Air Force- 03

Review Activities
Army- MD
Navy- MS
Air Force- 03

Preparing activity:
Defense Supply Agency.-DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

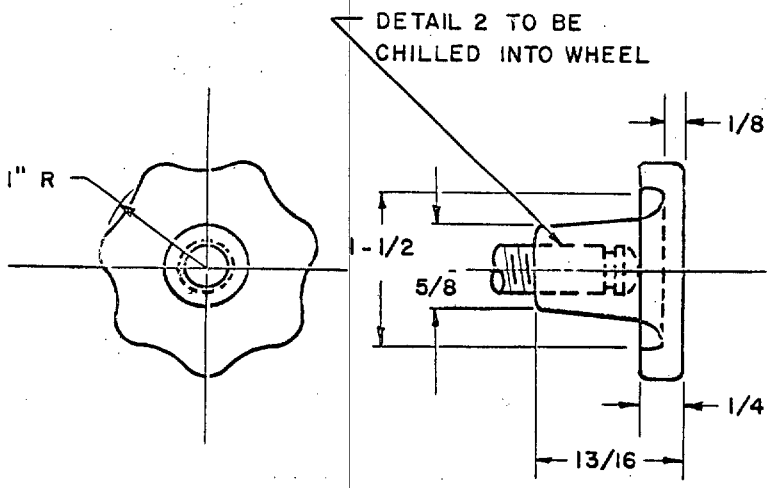
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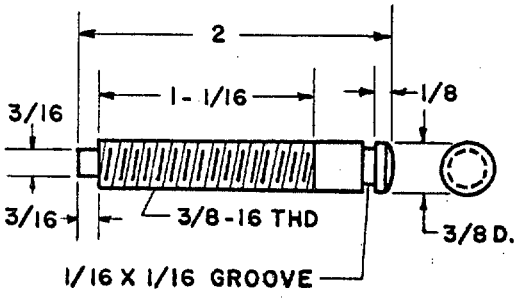
CLAMP PULLEY, LOOSE SWIVEL

Figure 2

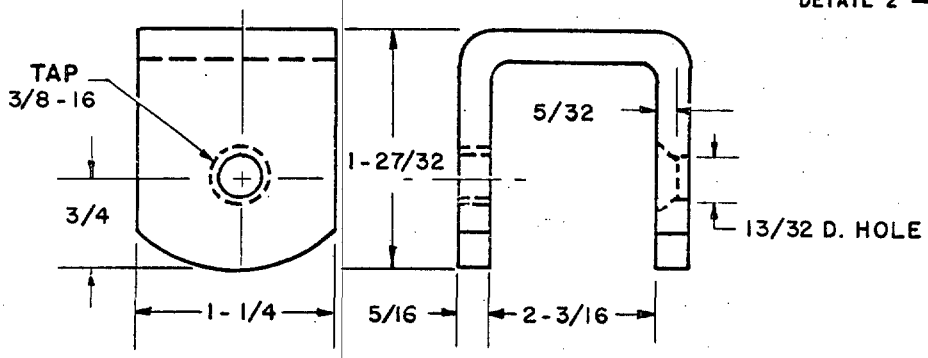
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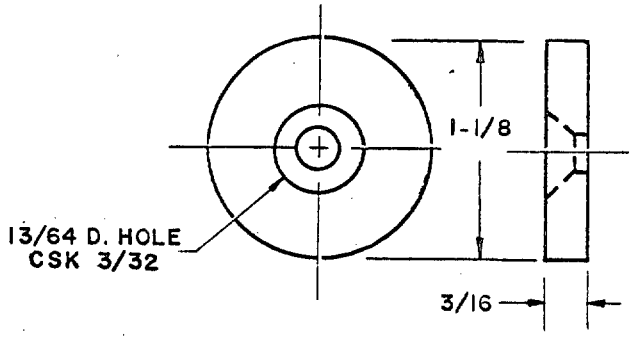
DETAIL 1 — CLAMP WHEEL



DETAIL 2 — CLAMP STUD



DETAIL 3 — "C" CLAMP

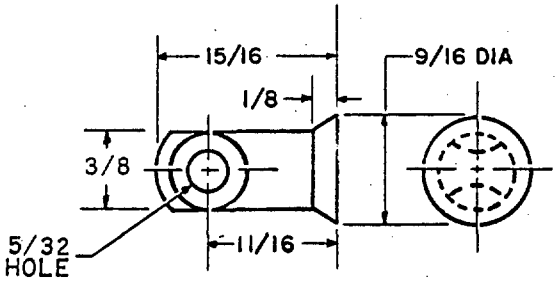


DETAIL 4 — CLAMP PLATE

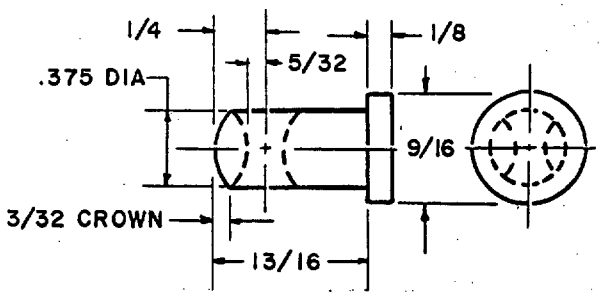
ALL MEASUREMENTS ARE IN INCHES
FRACTIONAL TOLERANCES = ±1/32
DECIMAL TOLERANCES = ±.005

Figure 3

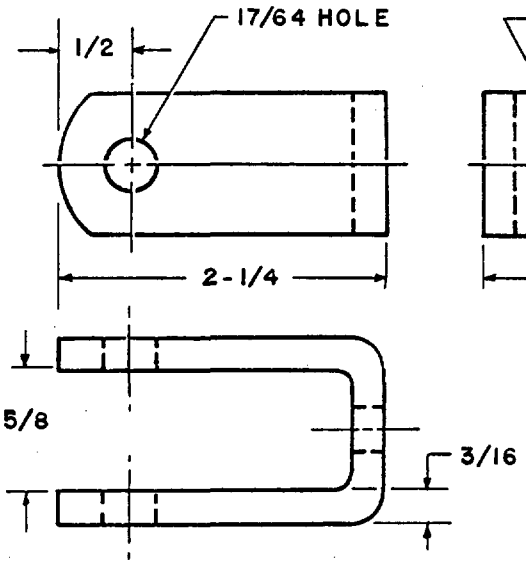
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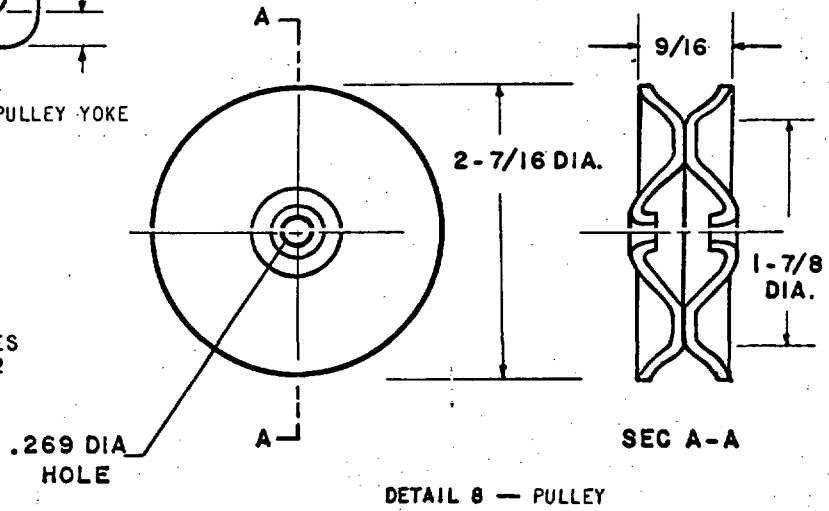
DETAIL 5 — SWIVEL PIN



DETAIL 6 — PULLEY YOKE, SWIVEL PIN



DETAIL 7 — PULLEY YOKE



DETAIL 8 — PULLEY

ALL MEASUREMENTS ARE IN INCHES
FRACTIONAL TOLERANCES = ±1/32
DECIMAL TOLERANCES = ±.005

Figure 1

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